

SOC Infrastructure Phase 1: Hybrid Environment Deployment

Project: Cross-Platform SIEM/XDR Deployment (LAN, Cloud & IoT)

Administrator: Cybersecurity Enthusiast | IT Specialist

Status: Phase 1 Complete - Fully Operational 

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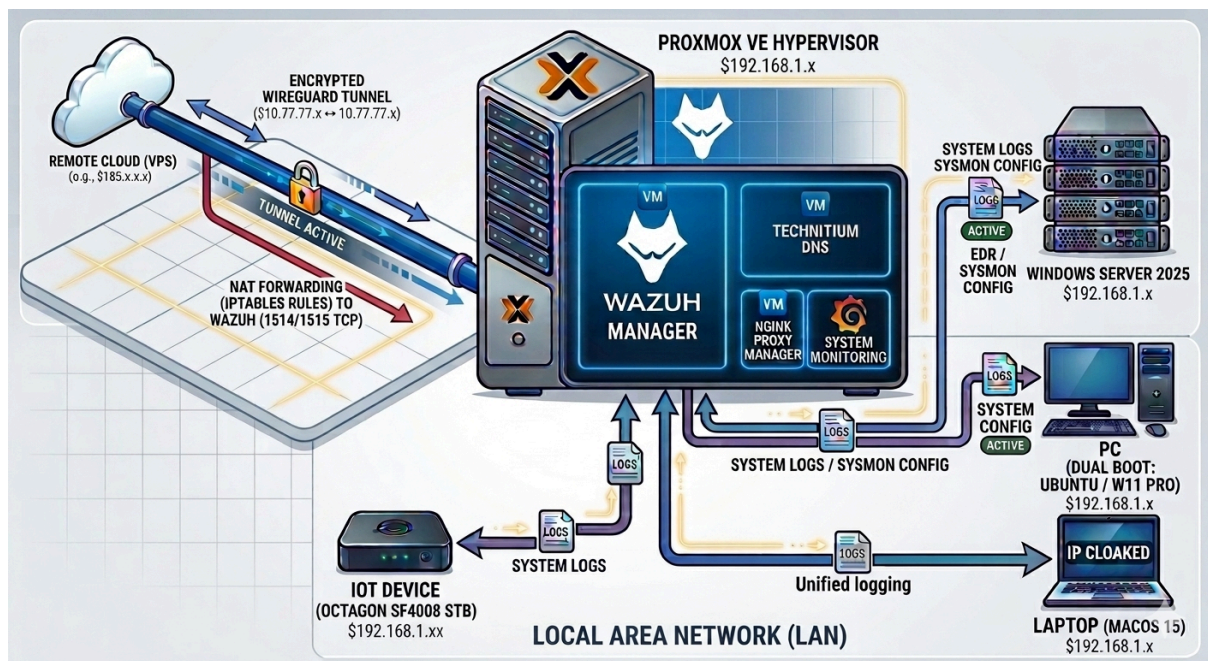
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1. Executive Summary

The objective of Phase 1 was to establish a high-fidelity monitoring ecosystem integrating local virtualization (**Proxmox**), cloud resources (**VPS**), and diverse endpoints (**Windows EDR, macOS, IoT**). By implementing a **Wazuh-centric SOC**, we have achieved 360-degree visibility across the network, ensuring that every system from the Windows Server to the Enigma2-based Octagon is a synchronized sensor for threat detection.

Network diagram



2. Core Infrastructure & Network Topology

A. Central Management Stack

- **Wazuh Manager:** Debian 13 (192.168.x.x) hosted on Proxmox.
- **Required Ports:** 1514/TCP (Event Data) and 1515/TCP (Agent Registration).

B. Cloud Connectivity (VPS over WireGuard)

The remote VPS (10.77.77.xx) is integrated into the SOC via an encrypted WireGuard tunnel. Direct communication is established between the VPS and the Proxmox Gateway (10.77.77.xx).

L3 Forwarding (NAT) Implementation:

To bridge the tunnel with the local Wazuh Manager, specific `iptables` rules were implemented on the Proxmox host:

Bash

```
# Forwarding Event and Registration traffic
```

```
iptables -t nat -A PREROUTING -i wg-overlay -p tcp --dport 1514 -j DNAT --to-destination 192.168.x.x:1514
```

```
iptables -t nat -A PREROUTING -i wg-overlay -p tcp --dport 1515 -j DNAT --to-destination 192.168.x.x:1515
```

```
# Masquerade for return traffic
```

```
iptables -t nat -A POSTROUTING -d 192.168.x.x -p tcp -j MASQUERADE
```

Strategic Analysis (Value Added):

- **Architectural Integrity:** By using `DNAT`, the Wazuh Manager's real LAN IP is never exposed to the external VPS. The VPS only "sees" the tunnel gateway (10.77.77.x). This adds a layer of **Security through Obscurity**.
 - **Routing Stability:** The `MASQUERADE` rule acts as the "Return Ticket" for data packets. Without it, the Manager would try to reply via the local default gateway instead of the tunnel, causing a **Routing Loop**.
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3. Advanced Endpoint Telemetry (Windows EDR)

Both Windows 11 and Windows Server 2025 have been upgraded to **EDR (Endpoint Detection and Response)** status through the integration of **Microsoft Sysmon**.

A. Sysmon Deployment & Configuration

Installation via PowerShell:

PowerShell

```
# Download and install Sysmon with a security-focused configuration
```

```
.\Sysmon64.exe -i sysmonconfig-export.xml -accepteula
```

B. Wazuh Ingestion (ossec.conf)

The agent configuration was modified to monitor the deep-level event channel:

XML

<localfile>

<location>Microsoft-Windows-Sysmon/Operational</location>

<log_format>eventchannel</log_format>

</localfile>

Strategic Analysis (Value Added):

- **Process GUID Tracking:** Standard PIDs are recycled by Windows; Sysmon GUIDs are unique. This allows the SOC to track a malicious process even if it renames itself or restarts.
 - **Lateral Movement Visibility:** By monitoring Network Connections (Event ID 3), we can detect if an attacker attempts to "jump" from a workstation to the Server.
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4. IoT Deep Dive: Octagon SF4008 (Enigma2)

The Octagon SF4008 (192.168.x.x) integration focuses on monitoring system telemetry directly within the LAN.

- **Log Source:** `/media/hdd/log/system.log` (Cleaned telemetry).
- **Resource Management: File Integrity Monitoring (FIM)** is restricted to `/etc` and `/usr/bin`.

Strategic Analysis (Value Added):

- **Zero-Latency Monitoring:** By keeping log delivery within the LAN, we ensure security tasks don't compete with the device's primary video processing.
 - **CPU Preservation:** Limiting FIM prevents the agent from "thrashing" the STB's older 4.1 Linux kernel, ensuring device stability.
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5. Comprehensive Troubleshooting (Network Forensics)

I. Handling "Ghost" Connections

- **The Issue:** Agents failed to connect due to stale tunnel addresses (10.77.77.x) lacking proper return routing.
- **The Fix:** Identified and deleted legacy `iptables` entries pointing to the old IP (192.168.30.x).

- **Analysis:** This demonstrated the importance of **Clean State Transitions**. Stale routes are a leading cause of "silent failures" in complex SOC environments.

II. Stale Keys & Identity Reset

- **The Issue:** Cryptographic key mismatches from previous installations.
- **The Fix:**
 1. **Manager:** Removed stale ID via `/var/ossec/bin/manage_agents -r [ID]`.
 2. **Agent:** Deleted local identity at `/var/ossec/etc/client.keys`.
 3. **Restart:** Triggered a fresh handshake by restarting the agent service.
- **Analysis:** This enforced the **Zero Trust** principle, ensuring every agent has a fresh, valid cryptographic handshake with the Manager.

6. Summary: Phase 1 Completion

Phase 1 has been successfully concluded. The SOC environment is now receiving consistent, high-fidelity telemetry from all endpoints:

- **Windows 11 & Server 2025:** Deep process and network analysis via **Sysmon**.
- **Remote Ubuntu VPS:** Securely integrated via **WireGuard** and NAT forwarding.
- **Octagon SF4008:** Specialized IoT monitoring for OSCam services.
- **Linux Nodes:** Full system visibility across Debian and Ubuntu instances.

